

Date: 21/12/2025

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Semester : BSAT 7th-Sem (Mor)
Subject : Parallel & Dist Computing

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ASSIGNMENT # 03

1. CPU Vs CUDA Performance Analysis

Problem Chosen:

Vector Addition:

$$C[i] = A[i] + B[i]$$

Implementation:

CPU: Sequential loop

GPU: Parallel kernel, one thread per element

Performance Comparison:

Aspect	CPU	GPU
Execution model	Sequential	Massively parallel
Time (large N)	High	Very Low
Scalability	Poor	Excellent
Throughput	Low	High

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2. CUDA Memory Analysis:

Global Memory:

Large, high latency
Used to store input/output vectors

Shared Memory:

On-chip, low latency
Not required for simple vector addition
Used in complex operations

Memory Access Patterns:

~~Coalesced~~ Coalesced global memory access
Results in maximum bandwidth utilization

Block & Grid Dimensions:

Optional block size: 256 or 512 threads
More blocks $\rightarrow$ better GPU occupancy
Poor sizing $\rightarrow$ idle cores and lower performance

3. CUDA Performance Bottlenecks

Major Bottlenecks:

Kernel Launch Overhead: Significant for small data sizes.

Host-Device Memory Transfer: PCIe transfer ~~layer~~ latency is expensive.

Thread Divergence: Minimal in vector addition.

Occupancy: Affected by block size and register usage.

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4. Distributed System Failure Analysis

Failure Types

Failure	Description
Node Failure	Entire machine crashes
Process Failure	Application crashes
Network Failure	Packet loss/delay

Impact

Incorrect results
 Increased latency
 Reduced throughput

Failure Classification

Type	Description
Fail-Stop	Node halts and is detected
Omission	Message loss
Byzantine	Arbitrary/malicious behavior

Qs. Fault Tolerance Mechanisms Comparison

Checkpointing Vs Replication:

Aspect	Checkpointing	Replication
Resource Cost	Low	High
Recovery Time	Slow	Fast
Runtime Overhead	Moderate	High

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Active vs Passive Replication

Aspect	Active	Passive
Execution	All replicates	One primary
Latency	Higher	Lower
Fault Recovery	Instant	Slower

6. Performance vs Reliability Trade-off

Effect of Fault Tolerance

Increases Latency

Reduces throughput

Improves availability and correctness

When ~~minimum~~ Minimal Fault Tolerance is Accepted

Real-Time Systems

Non critical batch processing

Performance-sensitive applications.